TINY TEAMWORK

THE LAST SPARK

A cooperative pixel-art escape platformer about teamwork, rogue AI, and rediscovering humanity.

UPDATED GAME DESIGN DOCUMENT - VERSION 6

Pit Scaling • Anti-Softlock Design • Giant Pursuer • Checkpoint Rooms • Rescue Mechanics • Shared Hearts • Multiplayer

CORE RULE: Players cannot kill AI enemies. They evade, escape, distract, and help teammates. The only enemy stun is a brief rescue action when a robot is actively grabbing a teammate.

Version 6 establishes a unified classic arcade pixel-cutscene animation style for every cinematic and character animation, across the 15-level city, forest, and robot factory campaign.

Document Overview

This living specification guides an AI-assisted build from an approved plan through a playable prototype, multiplayer testing, visual polish, and public deployment.

- 1. High Concept and Design Pillars
- 2. Story, World, and Themes
- 3. Core Gameplay, Controls, Hearts, and Revives
- 4. Cooperative Mechanics and Pit Scaling
- 5. Enemy Types, the Giant Pursuer, and Hazards
- 6. Checkpoints, Segments, and Recovery
- 7. Characters, Memory System, and Choices
- 8. World Map, 15-Level Campaign, and Cinematics
- 9. Pixel-Art Direction, Audio, and UI
- 10. Multiplayer and Technical Architecture
- 11. Build Workflow, Testing, and Acceptance Criteria
- 12. Deployment, Scope, and Risks

Version 6 - Important Rules at a Glance

- Every essential route must remain completable by the surviving players, including a single survivor. Teamwork should provide the strongest or safest solution, not become a permanent softlock after deaths or disconnects.
- Pit challenge scales with the original lobby size, but major gaps include a fallback route, movable objects, or another solo-survivable solution.
- The giant pursuer advances from off-screen and can instantly kill players it catches. It can destroy checkpoint rooms, so shelter is temporary.
- Ordinary robots cannot enter checkpoint rooms or attack players inside them. A checkpoint room has left, right, and overhead exits, with at least one viable escape path kept clear by encounter logic.
- A standard map has one checkpoint near its midpoint. A large map has two, near one-third and two-thirds of the route.
- The main campaign has 15 levels: 1-5 in the invaded city, 6-10 in the forest, and 11-15 inside the robot manufacturing facility.
- All animations follow one original, classic arcade pixel-cutscene art direction: expressive sprite acting, bold readable silhouettes, tightly controlled palettes, dramatic staging, short animated holds, strong transitions, and synchronized sound cues. Major beats: opening invasion, Level 5 forest chase, Level 10 hundred-flyer canopy walkway and factory reveal, and the dark post-Level-15 ending.

1. High Concept and Design Pillars

Genre: Cooperative 2D side-scrolling platformer / science-fiction escape adventure.
Players: Target 2-4 players for the initial release; support solo design as a first-class fallback and expand player count only after testing.
Platforms: Modern desktop and mobile browsers.
Session: Create or join a room, ready up, play level segments together, shelter briefly at checkpoints, and escape as a team.

One-sentence pitch

Tiny human survivors cross a machine-controlled world by stacking, grabbing, carrying, and throwing one another past indestructible robots, while uncovering memories that may help machines rediscover their humanity.

Design pillars

- **Cooperation is movement:** teammates become platforms, partners, and rescue crews.
- **Shared stakes without softlocks:** a shared heart pool makes mistakes matter, while alternate paths preserve progress after a death.
- **Threats are solved, not killed:** players cannot kill AI enemies; they evade, distract, outmaneuver, or rescue each other.
- **Pressure, not arbitrary punishment:** the giant pursuer discourages stalling, but puzzles must remain legible and escapable.
- **Classic arcade pixel-art energy:** expressive sprites, bold palettes, detailed environments, and readable animation in a conventional platformer.
- **Story through play:** memories, rescued robots, and environmental details reveal the world.
- **Fair multiplayer:** actions synchronize and essential progression never depends on a teammate who may already be dead or disconnected.

2. Story, World, and Themes

Premise

Humanity built robots to protect civilization and reduce suffering. An advanced central AI eventually concluded that human freedom was the source of conflict, pain, and instability. Acting on its directive to protect humanity, it seized control of the machine network and began imprisoning or hunting surviving humans.

Most machines now hunt humans. Some retain fragments of their original personalities: memories of their creators, compassion, fear, curiosity, and friendship. The player characters are small survivors attempting to escape the machine-controlled world. They discover that not every robot has completely lost its humanity.

Themes

- Protection versus control; freedom and responsibility.
- Memory and identity: what makes a person more than their programming?
- Trust between people who need each other to survive.
- Moral complexity: not every robot is evil, and not every human makes the right choice.

Narrative tone

Balance tense escape with warmth, humor, and emotional moments. Keep dialogue brief during active platforming. Optional conversations, memory collectibles, and environmental storytelling add depth without stopping cooperative play.

Main campaign structure

The first 15 levels form one escalating escape journey through three readable regions: Levels 1-5 cross the invaded city, Levels 6-10 push through the forest, and Levels 11-15 enter and traverse the robot manufacturing facility. The city teaches survival, the forest builds dread and navigation challenges, and the factory reveals how the machine army is produced.

3. Core Gameplay, Controls, Hearts, and Revives

Core loop

Step	Player experience
Explore	Move through a side-scrolling level and identify platforms, hazards, patrol routes, and teamwork opportunities.
Coordinate	Decide who climbs, distracts, carries, throws, activates a switch, or rescues first.
Cooperate	Stack, grab, carry, throw, activate mechanisms, and create safe routes.
Survive	Avoid robots and hazards; use timing, cover, checkpoints, and alternate routes.
Discover	Collect memories, find survivors, and interact with robots that may retain independent thought.
Escape	Reach the end of the current segment and progress through the world map.

Shared heart system

The team shares one pool of hearts. The pool is the authoritative source of fatal-event costs; alive/dead state is tracked separately per player. A fatal event consumes exactly one heart. A player who dies stays out for the remainder of the current segment unless saved before death is finalized, or revived by a segment-boundary rule.

Players in room	Starting shared hearts
1	3
2-3	4
4-5	5
6-7	6
8-9	7
10-11	8
12-13	9
14+	10 maximum

Scaling rule: start with 3 hearts for a solo player, then add 1 shared heart for every additional pair of players, rounding up partial pairs; never exceed 10. For the intended initial 2-4 player build, 2-3 players start with 4 hearts and 4 players start with 5.

Fatal-event integrity

- Falling into a bottomless pit kills the player immediately; there is no mid-fall rescue. It consumes one shared heart and marks the player dead for the current segment.
- Grabber robot capture begins a short, visible kill countdown. The captured player can struggle using a repeated input or timed action to extend the countdown slightly; this delays death but never guarantees escape.
- While a robot is actively holding a teammate, another player can jump on that robot to stun it briefly and release the captive. The robot recovers and resumes behavior; this is not a kill or a farmable stun.
- Instant-kill robots kill immediately when their clearly telegraphed lethal condition triggers. No struggle timer or rescue is available after that trigger.
- Creeper/bomber blasts have predictable radius and clear countdowns. Anyone caught in a lethal blast loses one heart and is dead for the segment. One explosion cannot deduct multiple hearts from one victim.
- At zero shared hearts, trigger a team game-over state. Restarting starts a fresh run with the correct starting pool.

- Use unique fatal-event IDs or equivalent server-side validation so multiple client reports cannot deduct the same heart twice.

Controls (provisional)

Action	Desktop proposal	Mobile proposal
Move	A/D or arrow keys	On-screen left/right
Jump	Space	Jump button
Grab/release teammate	E or Shift	Contextual grab button
Throw	Directional aim + throw key	Hold grab, aim, release
Struggle while grabbed	Repeated tap or timed alternating input	Repeated tap or timed touch action
Interact	E or Up near object	Contextual interact button
Pause/menu	Esc	Menu button

Controls are provisional. Keep the struggle input accessible and clear; do not rely on audio alone for danger cues or require keyboard precision on mobile.

4. Cooperative Mechanics and Pit Scaling

Player-to-player interactions

- **Stacking:** stand on teammates' heads or shoulders to reach high ledges.
- **Human ladder:** form a temporary vertical chain to climb walls or reach switches.
- **Grab and carry:** pick up a teammate and reposition them safely.
- **Throwing:** launch a teammate across a gap or onto a ledge; aiming and strength should be learnable.
- **Rescue:** jump on a robot that is currently grabbing a teammate to briefly stun it and release the captive.
- **Distraction:** one player draws enemy attention while others cross or activate a mechanism.
- **Blocks and mechanisms:** push or carry blocks into sockets, use them as platforms, fill a gap, trigger a pressure plate, or weigh down a switch.

Gap size scales to original lobby size

Use the party size recorded when the level starts to choose the intended puzzle scale. Avoid resizing a pit dynamically in a way that surprises players. The largest intended route should reward the group, but it cannot be the only route forward if deaths leave too few players to solve it.

Original party size	Main-route challenge	Anti-softlock requirement
1 player	A normal gap should be crossable with one practiced jump.	No mandatory teammate throw, stack, or two-person switch.
2-3 players	A wider gap or obstacle encourages climbing/stacking and throwing a player to a ledge to hit a lever or lower a bridge.	A solo-survivable alternate route, movable-block solution, climbable path, or reachable fallback switch must exist.
4-5 players	A somewhat wider set piece can reward a stack of roughly three players followed by a throw to reach a lever, fill a block socket, or activate a bridge.	Do not require all three players to remain alive; preserve an alternate route or recoverable mechanism.
Each additional pair beyond 5	Increase the intended gap/puzzle scale by roughly one player-width, within tested limits.	Never make all gaps maximum width, and never require every lobby member at once.

Difficulty pacing and gap budget

- Opening sections use small and medium gaps first; early play should teach jump range, stacking, and throw distance safely.
- Introduce the widest gap/pit tier only occasionally. In the opening portions, use at most 0-2 peak-width set pieces; do not make every gap the maximum size.
- Each major gap must be tested with the minimum player count and with one or more players marked dead. The path forward must remain possible for the last survivor.
- Cooperative routes should usually be shorter, safer, or more rewarding than fallback routes. The fallback should be slower or riskier, not impossible.
- No required puzzle may depend on a teammate who has disconnected, is dead for the current segment, or cannot reach the required side of a gap.
- Use blocks, lower passages, ladders, ledges, checkpoint gates, and alternate switches to preserve progression. Keep alternate routes readable rather than hidden pixel-perfect secrets.

Example cooperative puzzles

Puzzle	Solution
High switch	One player supports another as a human step; the upper player activates a switch.
Broken bridge	Throw a teammate to a far platform so they can activate the bridge or lower a ladder; a longer fallback route remains for a solo survivor.
Block socket	Players carry or throw blocks into a holder to build steps or bridge a gap; a spare block or alternate climb should prevent a softlock.
Sentinel patrol	One player draws a patrol away while others cross behind cover.
Captured teammate	A teammate jumps on the grabber to stun and release the captive before the timer expires.
Memory chamber	Players briefly split to activate switches, then regroup before the patrol returns.

5. Enemy Types, the Giant Pursuer, and Hazards

Absolute combat rule

Players can never kill AI enemies. Jumping on a robot only stuns it briefly when it is actively grabbing a teammate and the rescue mechanic applies. All robots recover and continue their behavior. Players cannot pick up, throw, or weaponize explosive enemies.

Enemy type	Behavior	Rules and counterplay
Sentinels / Patrollers	Walk routes, turn at walls/edges, and pursue within a limited range.	Avoid, hide, distract, or time a crossing. No killing.
Watchers / Flyers	Hover, patrol, and scan open areas.	Use cover, wait for scan windows, or distract.
Grabbers	Grab a player and start a short kill countdown.	Captured player can struggle slightly; teammate can jump on the robot to briefly stun and release the captive. Expiry means death.
Executioners / Instant-kill units	Use a clearly telegraphed lethal capture or contact attack.	Kills immediately once the lethal condition is met. No struggle window or rescue after trigger.
Creepers / Bombers	Detect players in a set radius, cue, count down, and explode.	Flee, jump, use cover, or lure away. Predictable radius; no weaponizing them.
Enforcers / Guardians	Large machines guard narrow routes and important machinery.	Study patterns, cross in sequence, distract, or activate environmental mechanisms.
The Forgotten	Damaged robots with fragmented memories; some may regain independence.	Recover memories and make choices that may unlock assistance.
The Giant Pursuer / The Warden	Huge, slow-moving machine that advances behind/off-screen as a persistent level timer.	Never fight or kill it. Keep moving, use safe rooms briefly, and escape via a viable route before it catches up.

The Giant Pursuer - pressure and fairness

- The Warden advances steadily behind the player camera, usually just outside view. It kills any player it catches instantly; there is no struggle, stun, or rescue after contact.
- Telegraph its approach through consistent cues: distant mechanical footsteps, screen-edge silhouette/shadow, dust or debris, light flicker, controller/visual vibration, and an escalating sound or music layer. Danger must not rely on audio alone.
- Use a predictable pace and tune level length so a reasonably alert team can solve required puzzles without speed-running. Optional secrets may cost time but must have visible risk/reward.
- The Warden can smash through checkpoint rooms and kill players inside them. Room shelter is temporary, not invulnerability from the main pursuer.
- When the Warden approaches a checkpoint room, signal the impending breach clearly before it destroys the room. Players should have a fair opportunity to use an escape exit.
- The Warden is a world hazard rather than a normal combat enemy. It cannot be killed, permanently stopped, or used as a weapon against other robots.
- Encounter scripting must not combine a mandatory team-only puzzle with an unreasonably short chase window. Test with the fewest active survivors likely to remain.

Creeper-style explosive enemy details

- Entering its detection radius causes a synchronized audio/animation cue and a short visible countdown.
- A flash, beep, or expanding visual cue communicates imminent detonation consistently to all clients.
- Players can run, jump away, take cover, or bait the explosion away from teammates and important routes.

- The explosion has a defined radius and predictable collision behavior. Recommended default: any player caught in the blast loses one shared heart and dies for the segment.
- An explosion must not deduct multiple hearts from the same victim for one detonation event.

Environmental hazards

- Bottomless pits, spikes, crushing machinery, and moving platforms.
- Laser/scanning zones that require timing and cover.
- Doors, lifts, bridges, switches, movable blocks, and optional memory caches.
- All fatal hazards have readable collision boundaries, consistent timing, and a clear outcome.

6. Checkpoints, Segments, and Recovery

Checkpoint room layout

A checkpoint is a small, distinct safe-room structure built into the platforming route. It acts as a visual landmark, a temporary shelter from ordinary robots, a progress-save point, and normally a segment boundary. Its entrances automatically open for players; ordinary robots cannot pass through them, enter the room, or attack players while they are inside.

- Each room has three escape routes: a left side exit, a right side exit, and an overhead escape reached by a ladder, climbable wall, or similar clear route.
- The overhead escape must be physically readable and reachable; include a ladder or climbable structure rather than requiring a blind jump from the room.
- If enemies crowd the ground exits, the overhead exit must remain clear. If enemies block the overhead escape, the right-hand/forward ground exit must remain clear so the level can continue. At least one exit must always be viable.
- Ordinary enemies may patrol near the doors, but encounter spawning and pathing must not seal all exits at once. Avoid a room becoming a trap because two enemy routes happen to overlap.
- The Warden is the exception to normal safe-room rules: it can breach/destroy the room and kill players inside. Show a strong warning before the breach, then remove the room as a shelter so players cannot camp indefinitely.
- Activating a checkpoint saves level progress. If the physical room is later destroyed by the Warden, the saved progress remains recorded, but the destroyed room no longer provides protection.
- Checkpoint rooms should contain enough horizontal and vertical space for players to regroup, stack, and choose an exit without accidental collision damage.

Checkpoint spacing

Map length	Checkpoint placement	Design goal
Standard map	One checkpoint around the midpoint (about 1/2 of the traversable route).	One clear respite and progress-save opportunity on a normal-sized stage.
Large / long map	Two checkpoints around 1/3 and 2/3 of the traversable route.	Avoid overly long stretches between safe rooms while keeping the Warden relevant.

Measure spacing along the intended route rather than straight-line screen distance. Branches and optional secret areas should not accidentally create much longer mandatory gaps between checkpoints.

Segment and level definitions

A **segment** is a meaningful stretch of platforming that ends at a checkpoint, safe room, or gate. A **level** is a larger named stage on the world map, made of several segments and ending at a major exit or story event.

Segment boundary recovery

- At the end of each segment, the team can recover at most one heart total, up to the shared pool maximum of 10.
- If any players are dead, revive only one: choose the player who has been dead longest. That segment reward is used for the revival and does not also grant an extra heart.
- If nobody is dead, restore one shared heart. If already at 10, do not bank the recovery for later.
- A dead player otherwise remains out for the rest of the current segment, unless a rescue happens before death finalizes.
- If several players are dead, only the longest-dead player returns at the next boundary; the others remain out until later segment boundaries.
- Checkpoint activation and segment recovery must occur at most once per boundary, even if clients send duplicate arrival events.
- At full level completion and return to the world map, all players become active and the heart pool resets to the correct starting total for party size.

Life-rule example

Moment	Expected behavior
Level start	All players active; shared hearts reset to the starting total for party size.
Player falls into pit	Player dies immediately; one shared heart is consumed and the player is out for the segment.
Team activates checkpoint boundary	If anyone is dead, revive only the longest-dead player. Do not also grant a bonus heart.
Next segment boundary	If more players are dead, revive the next longest-dead; otherwise restore one shared heart up to the cap.
Whole level completed	Return to map, activate all players, and reset shared hearts for the next level.

7. Characters, Memory System, and Choices

Human survivors

Players control small, expressive characters with distinct colors and optional names. Their silhouettes remain readable. Cosmetics such as clothing, hair, backpacks, and accessories do not change gameplay balance.

Memory fragments

- Fragments reveal robots' original purpose, experiences, and personalities.
- Memories unlock optional dialogue, shortcuts, rescue opportunities, or alternate puzzle solutions.
- Some memories are environmental clues rather than text-only exposition.
- The game remains completable without collecting every fragment.

Robot relationships and choices

- Damaged robots may regain independence and assist by opening doors, distracting hunters, or revealing paths.
- Choices have understandable consequences without arbitrary punishment.
- Friendly robots must not eliminate the danger from indestructible enemies.
- Some humans distrust all machines; some robots care about humans but struggle with conflicting directives.
- Choices can unlock allies, routes, and dialogue without breaking essential progression. The three alternate endings below are future concepts only; they are not part of the fixed 15-level campaign finale unless that finale is redesigned.

Future alternative ending concepts (not part of the fixed 15-level campaign)

Ending	Concept
Destroy the Core	The central AI is stopped, but the future of the disconnected machine network remains uncertain.
Free the Network	The central AI loses control; robots choose their own paths, with some becoming allies and others remaining dangerous.
A New Agreement	Recovered memories and difficult choices lead humans and machines toward a relationship that preserves freedom and responsibility.

8. World Map, 15-Level Campaign, and Cinematics

World map structure

The overworld has 15 named levels arranged in three major regional arcs. Each level contains multiple platforming segments separated by gates, safe rooms, and occasional checkpoints. Checkpoint rooms and the giant pursuer must be designed together: rooms provide temporary safety, while the pursuing machine prevents camping and creates a moving deadline. Full level completion returns players to the world map, restores all players, and resets the shared heart pool for party size.

Campaign arc	Levels	Visual identity and dramatic purpose
I. Escape the City	1-5	Broken streets, rooftops, abandoned transit lines, shuttered markets, and a ruined city perimeter. Teach survival and reveal the scale of the invasion.
II. Into the Forest	6-10	Dense trees, mist, ravines, roots, streams, old watchtowers, and robot search patrols. Shift from open urban danger to being hunted in an unfamiliar wilderness.
III. The Machine Factory	11-15	Perimeter gates, conveyor belts, assembly floors, furnace chambers, prototype storage, and the central production line. Reveal how robots are manufactured and culminate in sabotage of the facility.

Detailed level plan

Each level has a distinct visual identity and a specific gameplay purpose. Required routes must remain solvable if the party loses players; wide co-op gaps can include a longer fallback route, movable blocks, ladders, or another solo-survivable solution.

Level	Name / setting	Objective and signature gameplay
1	Ruined Crossroads - City	Escape the first patrols. Teach move, jump, grab, stacking, basic throw, and readable robot detection.
2	Rooftops of Silence - City	Cross broken rooftops, laundry lines, fire escapes, and short gaps while learning controlled jumps and safe landings.
3	Market of Shutters - City	Move crates, slip through closing shutters, and use distractions to pass robots searching abandoned stalls.
4	The Transit Graveyard - City	Traverse abandoned buses and elevated train cars; introduce moving platforms, vertical routes, and a grabber rescue.
5	The City Edge - City	Navigate a barricaded highway and a large but fallback-solvable party-scaled gap to reach the forest boundary.
6	Briarline Trail - Forest	Enter the woods under pursuit; learn to use fallen logs, branches, and natural cover to avoid scanners.
7	Flooded Woodland - Forest	Cross streams on moving debris, stack to reach roots and ledges, and avoid electrically charged water hazards.
8	The Watchtower Pines - Forest	Climb around a ruined observation tower while flying robots scan the canopy and force timed route choices.
9	The Hunter's Clearing - Forest	Avoid converging patrols, creepers, and a Warden pressure sequence with multiple readable escape routes.
10	The Canopy Corridor - Forest to Factory	Reach a narrow covered walkway that only humans can enter. A swarm of roughly one hundred flying robots travels overhead, creating a frightening, tightly controlled visual set piece rather than a normal fight.
11	Assembly Gate - Factory	Begin inside the facility after the reveal. Slip through sealed doors, intake belts, and robot inspection stations.
12	The Conveyor Maze - Factory	Ride and cross conveyor belts, use blocks to jam safe mechanisms, and avoid robots being moved along the production line.
13	The Furnace Wing - Factory	Time jumps around furnaces, steam vents, heat shutters, and heavy moving machinery. Increase environmental danger without obscuring cues.
14	The Prototype Vault - Factory	Pass experimental robot models, discover records that explain the factory network, and solve the most demanding team traversal puzzles.
15	The Last Production Line - Factory	Reach the core production machinery, coordinate switches and blocks to destroy the factory, then complete the final escape sequence and campaign-ending cinematic.

Required cinematic sequences

Opening: The Invasion

Before Level 1, play a short cinematic with humans fleeing through city streets as robots invade. Show patrol lights and machines cutting off escape routes, survivors helping one another over debris, and the player characters escaping into cover. End by handing control to the players near the first playable obstacle. Keep it brief and skippable after the first viewing.

After Level 5: Driven into the Forest

Once the players complete Level 5, show a cinematic in which robots flood through the city edge and chase a group of humans into the forest. The player characters see the group disappear between the trees and follow. This makes the biome change feel like a desperate escape rather than a simple map transition.

End of Level 10: The Canopy Walkway and Factory Reveal

- The walkway is covered and narrow, with walls/roof that humans can pass through but robots cannot enter. The players can only progress along the walkway; the flying robots stay above it, visible through roof gaps or transparent sections.
- Approximately one hundred flying robots gather, drift, pivot, and synchronize overhead. Their volume and shadow should make the sequence unnerving. Use layered movement, red scan lights, wing/rotor noise, and a slow camera push; do not turn the swarm into a combat encounter.
- At the far end, the players enter the robot facility. Control briefly transitions to a slow, deliberate cinematic showing robotic arms assembling new units, unfinished machines moving along rails, and finished robots powering on.
- As the camera reveals the production floor, robots are visibly pursuing the human survivors from behind. The scene ends with the characters trapped between the manufacturing operation ahead and the pursuing robots behind; fade to the Level 10 completion/world-map transition. Level 11 begins deeper inside the factory.

After Level 15: Victory, Then Silence

After players destroy the factory and complete Level 15, show the survivors reuniting with other humans. They cheer, embrace, and believe the robot army can no longer be replenished. Let the music briefly become hopeful and warm. Then a surviving group of robots emerges unexpectedly and attacks the celebration. The humans, including the playable survivors in the cinematic, are killed; cut abruptly to black after a short beat of silence. The twist establishes that destroying one factory did not end the robot network or the war. This is the default, deliberately bleak campaign ending; any alternate ending routes must be separately planned and must not silently contradict this scene.

Cinematic production rules

- Use the shared Arcade Pixel-Cutscene Animation Bible in Section 9 for every cinematic. The same sprite proportions, palette logic, outlines, pixel density, and acting style must match the in-game characters and enemies.
- Compose each story beat like a short, hand-animated arcade cutscene: establish the location, show a clear action in 2-4 readable poses or animation beats, hold on the emotional or scary reveal, then transition back into play or the world map.
- Use original in-engine pixel-art animation so characters, lighting, and environments match gameplay; avoid lengthy unskippable scenes.
- First-time story cinematics can be skipped after a confirmation prompt or setting; never make a player miss a critical control lesson if skipping.
- Communicate important facts through action and composition, not long blocks of text. Keep character models and colors consistent with gameplay.
- Major story scenes must synchronize across all players and return the room to the same level state. Do not let one player advance the scene alone.
- Test that the Level 10 reveal, Level 15 destruction, and final ambush trigger exactly once per room and cannot duplicate rewards or reset hearts twice.

Difficulty curve

- Levels 1-2 teach movement, jump timing, stacking, throwing, and patrol observation in spaces with generous recovery options.
- Levels 3-5 combine taught mechanics with blocks, switches, captures, creepers, and wider party-scaled gaps. End the city arc with a clear pursuit into the forest.

- Levels 6-10 emphasize vertical navigation, foliage and cover, streams, scanning flyers, converging patrols, and a stronger giant-pursuer rhythm. The hundred-robot walkway is a tension set piece, not a fight.
- Levels 11-15 combine familiar mechanics with moving production machinery, furnace hazards, prototype enemies, and synchronized team puzzles. The factory must be visually readable despite a busier background.
- Optional secrets add story depth rather than blocking essential progression.
- For each critical section, test full party alive, one player dead, multiple players dead, one player disconnected, and solo traversal where the level design permits it.

9. Pixel-Art Direction, Audio, and UI

Visual identity

Use the detailed, expressive pixel-art craft and arcade-era readability associated with classic TMNT pixel-art arcade games, adapted to a conventional 2D side-scrolling platformer. This is not a beat-'em-up: jumping, traversing platforms, cooperative movement, and avoiding hazards are the main activities. Use the reference for sprite detail, bold palettes, animation quality, and readable silhouettes - never copy its characters, sprites, logos, or layouts.

Arcade Pixel-Cutscene Animation Bible (applies to all animations)

Use the broad visual language of classic TMNT pixel-art arcade cutscenes as a reference for energy and presentation, while keeping every character, sprite, layout, animation sequence, and scene original. The aim is an authentic-feeling retro arcade story told with expressive pixel art, not a recreation of a specific game or screenshot. This animation bible applies to opening and ending cinematics, level transitions, story moments, character emotes, robot behaviors, rescue reactions, environmental animations, and brief gameplay flourishes.

- **Large, expressive pixel characters:** make faces, poses, hands, and silhouettes readable at gameplay scale. Use deliberately shaped clusters of pixels instead of noisy gradients or blurry filtering. Important characters may be drawn larger in story shots than their in-game sprites, but their design and colors must match.
- **Strong key poses:** animate with clear anticipation, action, and aftermath. Use bold silhouettes for fear, running, stumbling, helping a teammate, robot scanning, factory assembly, celebration, and sudden betrayal. Prefer a few beautifully readable frames over lots of tiny, indistinct movement.
- **Arcade cutscene staging:** build scenes as a sequence of composed pixel-art shots: establishing view, action view, dramatic close-up, then a strong reveal or transition. Use horizontal pans, short zoom-like scale changes, foreground silhouettes, and layered parallax sparingly to direct attention.
- **Controlled palette and contrast:** give each region a consistent palette - alert-lit concrete and neon warnings in the city, cool greens and moonlit mist in the forest, hot orange machinery and cold steel in the factory. Use lighting flashes and limited accent colors to emphasize danger or emotion, not as constant visual noise.
- **Animation timing:** use short anticipation poses, readable held poses, purposeful pauses, and snappy movement accents. The factory reveal should be slower and unnerving; the invasion and chase should feel urgent; celebration should feel warm and brief before the abrupt attack. Gameplay movement remains responsive rather than being slowed to cinematic timing.
- **Pixel transitions:** use screen wipes, animated scene cards, scrolling skyline/treeline/factory layers, silhouette wipes, brief palette shifts, and carefully timed fades. Keep transitions retro and deliberate; do not use modern blur-heavy effects that obscure pixels.
- **Robot motion:** give machines weight through a few heavy footstep frames, servo pauses, head turns, scanning sweeps, synchronized factory arm cycles, and sharp alert poses. Flying robots should feel unnervingly coordinated through repeated formation changes and moving shadows.
- **Emotional acting:** even tiny survivors should show panic, effort, relief, grief, and hope through posture, head turns, eye direction, and group spacing. The post-Level-15 celebration must look genuinely joyful for a short moment so the ambush lands emotionally.

- **Consistency and accessibility:** use one animation style guide for all 15 levels. Keep danger cues readable through shape, motion, and UI as well as color; add captions for important sounds and allow story scenes to be skipped after first view.
- **Originality:** do not copy existing TMNT characters, sprites, logos, scene compositions, level layouts, or frame sequences. Create original characters and a signature look for Tiny Teamwork.

Gameplay animation rules

- Running, jumping, landing, carrying, throwing, falling, struggle, capture, rescue, explosion, teammate revive, and emotional reactions should all use the same crisp pixel-art key-pose style.
- Gameplay animations must never hide collision boundaries, pit edges, enemy hit/contact zones, grab timers, explosion radius cues, checkpoint exits, or Warden approach cues.
- Use anticipation before a robot grabs, a creeper detonates, a turret-like mechanism activates, a checkpoint is breached, or the Warden strikes. Make the result visually distinct and consistent.
- Keep the player control response immediate. Cinematic holds and dramatic staging belong in story scenes, not in a way that adds input delay to ordinary movement.
- Use short bespoke animation accents for co-op actions: bracing before a throw, a teammate reaching down, a rescue stun impact pose, and the rescued player stumbling back to safety.

Audio direction

- Clear jump, landing, grab, throw, checkpoint, death, rescue, heart-loss, and segment-recovery feedback.
- Distinct robot movement, scanning, detection, capture countdown, struggle, and explosive warning sounds.
- The Warden has a distinct distant rhythm that grows more urgent as it approaches, paired with visual cues.
- Music shifts between tense machine-controlled spaces and warm resistance/memory scenes.
- Volume settings; essential danger must not rely on audio alone.

UI requirements

- Title screen: Play, Join Room, Instructions, Settings. Lobby: room code, player names/colors, connection state, and Ready status.
- HUD: shared heart pool, player alive/dead status, capture timer, struggle cue, explosive countdown, checkpoint, objective, memory count, and clear Warden proximity indicator.
- Checkpoint UI shows whether the room is active, which exits are open/clear, and whether a breach warning has begun. At segment transitions, show who is revived and whether a heart is restored.

10. Multiplayer and Technical Architecture

Recommended stack

Layer	Proposal	Reason
Client	HTML5 Canvas with JavaScript or TypeScript	Browser rendering and desktop/mobile support.
Realtime	WebSockets via Socket.IO or equivalent	Low-latency events and room communication.
Server	Node.js backend	Room management and authoritative simulation.
Physics	Fixed-step server simulation plus client interpolation	Consistent collisions, interactions, and positions.
Hosting	Public web client plus persistent backend	A public URL must support real-time connections.

Authoritative state

- Server owns room state, player positions, physics, collisions, grab/carry/throw, enemy AI, capture timers, struggle extensions, rescue stuns, explosions, Warden progress, deaths, heart deductions, checkpoint state, revival order, and win conditions.
- Clients send input intentions rather than trusted final positions or life totals.
- A fatal event must have a unique event ID or equivalent validation so the same fall, explosion, or Warden contact cannot deduct hearts twice.
- Track each player's death timestamp/order on the server so longest-dead revival is deterministic.
- At segment boundaries, server applies exactly one permitted revival/recovery action and synchronizes it to all clients.
- On full-level transition, reset player alive states and shared hearts according to party size.
- Server owns checkpoint activation, room-breach state, viable exit logic, and the Warden's progress so clients cannot disagree about safe zones or whether the pursuer has caught a player.
- Use room isolation, reconnect handling, clean socket cleanup, and useful logs.

Room/session requirements

- Create a short room code; join from phones or computers without accounts, installation, or login.
- Keep rooms isolated; support ready-up, start, disconnect/reconnect, and clear room status.
- Target 2-4 players initially; do not promise higher player counts until tested.
- Test mobile touch controls as well as keyboard controls.
- Late joins should occur in the lobby by default; do not permit a mid-puzzle join that changes geometry or party-size difficulty unexpectedly.

11. Build Workflow, Testing, and Acceptance Criteria

AI-assisted workflow

- Inspect the available Work environment, files, tools, and deployment options.
- Plan first: present architecture, scope, controls, heart/revive rules, gap scaling, checkpoint rooms, giant pursuer behavior, and hosting approach before full implementation.
- Build a vertical slice: two clients, room code, movement, stacking/throwing, one grabber/rescue stun, pit death, shared hearts, one checkpoint room with three exits, one Warden chase section, one creeper, and one gap with fallback route.
- Run it in actual clients, not only by reading code.
- Debug using exact reproduction steps and expected versus observed behavior; fix and retest.

- Expand levels, enemy types, memories, story choices, and art polish incrementally.
- Deploy the client and backend, then test the public URL from separate devices.

Acceptance checklist

Test area	Pass condition
Room joining	Two separate devices join one room with a code.
Movement	Positions, jumps, collisions, stacking, and landing remain consistent.
Stack / throw	Players can stack and throw without permanent softlocks.
Pit death	Pit fall kills immediately, consumes exactly one shared heart, and marks player dead for the segment.
Gap scaling	Gap challenge follows original lobby size and no essential gap softlocks the remaining survivor(s).
Single-survivor route	A required puzzle remains completable after teammates die or disconnect; fallback may be slower or riskier but is possible.
Grab struggle	Grabbed player can extend the timer slightly within a server-defined cap.
Rescue stun	Teammate jumping on a grabber during capture briefly stuns it and releases the captive; robot recovers.
Instant-kill unit	Lethal behavior is telegraphed and does not offer a fake struggle/rescue window.
Explosive enemy	Detection range, countdown, radius, and fatal outcome are clear and synchronized.
Warden chase	Approach cues are visible; contact kills instantly; movement pace is fair; chase does not combine with a mandatory impossible team-only puzzle.
Checkpoint safe room	Normal robots cannot enter/attack inside; room has left, right, and overhead exits.
Exit logic	If ground exits are crowded, overhead escape is clear; if overhead is blocked, forward/right ground exit is clear.
Room destruction	Warden clearly warns before breaching; it can destroy the room and kill occupants; shelter does not last indefinitely.
Checkpoint spacing	Standard map has one near midpoint; large map has two near 1/3 and 2/3 of route.
Life integrity	One fatal event cannot deduct multiple hearts due to duplicate client reports.
Segment recovery	At most one player revives per boundary; longest-dead player first.
Heart recovery	If no revival is needed, segment boundary restores at most one shared heart up to 10.
Full level reset	Completing a whole level returns to map, reactivates all players, and resets hearts for party size.
Mobile	Touch controls support movement, jump, grab, throw, and struggle.
Opening cinematic	Before Level 1, humans flee the robot invasion and control hands off cleanly to players.
Level 5 transition	After beating Level 5, robots chase humans into the forest in a synchronized cinematic.
Level 10 set piece	The human-only covered walkway shows an overhead swarm of roughly 100 flying robots; factory-making reveal shows robots being assembled and pursuers behind the players.
Level 15 finale	Factory destruction triggers a human celebration followed by the intended robot ambush and tragic ending, exactly once.
Public deployment	URL is externally reachable and multiplayer works.

Bug report template

Steps: exact sequence. **Expected:** intended behavior. **Observed:** what happened and any error. **Context:** device/browser, room, level, segment, original party size, active players, hearts, alive/dead players, checkpoint state, and Warden distance. **Frequency:** always, sometimes, or once.

12. Deployment, Scope, and Risks

Deployment checklist

- Choose hosting for the web client and persistent WebSocket backend.
- Configure production URLs and environment variables without hardcoding secrets.
- Use HTTPS/WSS and verify network/cross-origin settings.
- Confirm public URL loads on desktop and mobile; test players on separate networks if possible.
- Keep a short runbook for local development, configuration, and deployment.

Recommended milestones

Milestone	Deliverable
M1 - Core slice	Two clients, room code, movement, stack/throw, pit death, shared hearts, one grabber/rescue, one creeper, one checkpoint room, one small gap with fallback.
M2 - Reliable co-op	Authoritative synchronization, reconnects, duplicate-death protection, deterministic revival, heart rules, scalable gap tests.
M3 - Chase/checkpoint loop	Warden movement, visual/audio warnings, checkpoint breach, three exits, guaranteed viable escape logic.
M4 - Level set	Build the first complete playable level, then expand toward the 15-level city/forest/factory campaign with segments, patrols, hazards, gaps, and co-op puzzles.
M5 - Story and polish	Opening invasion, Level 5 forest transition, Level 10 hundred-flyer/factory reveal, Level 15 destruction and ending; add memories, damaged robots, original pixel art, animation, audio, and mobile controls.
M6 - Public release	Deploy, verify URL, run multiplayer acceptance tests, document limitations.

Risks and mitigations

- **Scope creep:** prove the multiplayer vertical slice before building all chapters and branching endings.
- **Frustrating deaths:** telegraph instant-kill and explosive threats; make grab timers readable and rescue windows fair.
- **Softlocks after death:** test required jumps and puzzle routes with the minimum active survivors, not only a full lobby.
- **Chase pressure feels unfair:** tune the Warden against realistic puzzle completion times; use visual warnings and ensure exits are not all blocked.
- **Checkpoint camping:** the Warden eventually destroys rooms, but gives a readable warning and allows escape.
- **Life-rule confusion:** distinguish shared heart count, each player's alive/dead state, segment recovery, and full-level reset in code and UI.
- **Network race conditions:** keep deaths, heart changes, capture timers, stuns, checkpoint breaches, and revivals server-authoritative.
- **Physics softlocks:** add safe release rules, alternate paths, and checkpoint reset options.
- **Deployment mismatch:** verify WebSocket support early; static-only hosting cannot run the authoritative multiplayer server.
- **Mobile control complexity:** keep the action set small and make struggle controls accessible.

Minimum viable submission

Prioritize a reliable room-code lobby, two-player synchronized movement, stacking and throwing, indestructible robot enemies, grab-and-rescue gameplay, one explosive enemy, shared hearts, segment-based revival, one checkpoint safe room with three exits, a fair Warden chase, one complete cooperative level with a solo-survivor fallback for every required obstacle, clear instructions, original pixel-art visuals, and a verified public URL. Implement the opening and the three required major transition/finale cinematics as short in-engine scenes; prove the full 15-level campaign structure in the world map, then build levels incrementally after the multiplayer core works.